

Department of Aerospace Engineering

JGI Global Campus, Jakkasandra Post, Kanakapura Taluk, Ramanagara District, Pin Code: 562 112
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End Year Project

Report on

“HIGH ALTITUDE LONG ENDURANCE (HALE) UNMANNED AERIAL VEHICLES (UAVs): SUSTAINED FLIGHT THROUGH RENEWABLE ENERGY”

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY IN AEROSPACE ENGINEERING

Submitted by

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CERTIFICATE

This is to certify that the project work titled “**HIGH ALTITUDE LONG ENDURANCE (HALE) UNMANNED AERIAL VEHICLES (UAVS): SUSTAINED FLIGHT THROUGH RENEWABLE ENERGY**” is carried out by **Parth (22BTRAS031)** a Bonafide students of Bachelor of Technology at the Faculty of Engineering & Technology, Jain (Deemed-to-be) University, Bangalore in partial fulfilment for the award of degree in Bachelor of Technology in Aerospace/Aeronautical Engineering, during the year **2025-2026**

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I, **Parth (22BTRAS031)** are students of 7th semester **B. Tech in Aerospace/Aeronautical Engineering**, at Faculty of Engineering & Technology, **JAIN (Deemed-to-be University)**, hereby declare that the End Year Project Report titled “**HIGH ALTITUDE LONG ENDURANCE (HALE) UNMANNED AERIAL VEHICLES (UAVs): SUSTAINED FLIGHT THROUGH RENEWABLE ENERGY**” has been carried out by me and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Aerospace/Aeronautical Engineering** during the academic year **2025-2026**. Further, the matter presented in the project has not been submitted previously by anybody for the award of any degree or any diploma to any other University, to the best of our knowledge and faith.

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Abstract

High Altitude Long Endurance (HALE) Unmanned Aerial Vehicles (UAVs) are designed to operate at altitudes above 18 km with mission durations extending from several days to multiple weeks. This capability is significantly higher and longer than conventional drones, which typically operate at lower altitudes and have shorter flight durations. The increasing demand for persistent surveillance, atmospheric monitoring, and communication relay platforms has accelerated interest in solar-powered and hybrid-energy UAVs capable of sustained flight. This project focuses on the conceptual design, performance analysis, and feasibility assessment of a solar-powered HALE UAV. The conceptual design phase involves creating a detailed plan for the UAV, considering factors such as its shape, size, and materials. Performance analysis involves evaluating how well the UAV will perform under various conditions, such as different weather patterns or payload weights. Feasibility assessment involves determining whether the UAV can be built and operated within practical constraints, technological limitations. To achieve these goals, the project involves a comprehensive literature review, aerodynamic analysis, and energy system assessment. The literature review involves examining existing research and technology to identify best practices and potential challenges. Aerodynamic evaluation involves studying how the UAV will interact with the air at high altitudes, which is crucial for ensuring stable and efficient flight, this evaluation will be carried out by simulating model with tools like ANSYS, NASTRAN. Energy system assessment involves evaluating the UAV's power sources and consumption to ensure it can sustain long-duration flights. The project addresses several key design challenges. One of these is low Reynolds number aerodynamics. At high altitudes, the air is thinner, which affects how it flows around the UAV. This can impact the UAV's stability and efficiency, so it's essential to design the UAV to perform well under these conditions. Another challenge is lightweight composite structures. To achieve long flight durations, the UAV must be as light as possible. However, it must also be strong and durable enough to withstand the harsh conditions at high altitudes. This requires the use lieteratur survey of advanced composite materials that offer a high strength-to-weight ratio. The third challenge is hybrid renewable energy integration. While solar power is a clean and abundant energy source, it's not always available, especially at night. Therefore, the UAV may need additional energy sources, such as batteries or fuel cells, to ensure continuous operation. To validate performance assumptions and analyze energy storage limitations, the project employs simulations and theoretical modeling. These tools allow researchers to test the UAV's performance under various conditions and identify potential issues. They can also help identify opportunities for future enhancements, such as improving the UAV's energy efficiency or increasing its payload capacity. The work conducted in this project contributes significantly to the understanding of long-endurance UAV concepts. It provides valuable insights into the design, performance, and feasibility of solar-powered HALE UAVs. Moreover, it lays the foundation for the entrepreneurial transformation of high-altitude UAV technology into viable commercial, defense, and civil solutions. Defense applications could include surveillance, reconnaissance, or communication relay. Civil applications could include environmental monitoring, disaster response, or search and rescue operations. In conclusion, the project on the conceptual design, performance analysis, and feasibility assessment of a solar-powered HALE UAV is a significant step forward in drone technology. It addresses critical design challenges and employs advanced tools and techniques to validate performance assumptions and identify opportunities for future enhancements. The insights gained from this project can pave the way for the development of innovative and practical applications of high-altitude UAV technology in various fields.

Table of Contents

Abstract	
1. Introduction	1
2. Market Survey and Literature review	2
3. Aim and Objectives	3
4. Problem Definition and Design Specifications	4
5. Methodology	5
6. Requirements / resources	9
7. Conclusion and Future Work	9
References	10

1. Introduction

HALE UAVs occupy a unique niche in the unmanned aircraft landscape, bridging the gap between conventional aircraft and satellites. Their ability to remain aloft for days or even weeks at altitudes above 18 km makes them indispensable for persistent surveillance, atmospheric research, and as low-orbit communication relays that can provide broadband connectivity to remote regions. The literature consistently reports that the most advanced platforms such as the NASA Helios-A, the Airbus Zephyr, and the Indian HAL Hansa have achieved endurance figures that exceed 24 hours, with some prototypes demonstrating multi-week sorties when powered by a combination of solar arrays, high-energy-density batteries, and, in a few cases, fuel cells. These endurance figures are not merely a function of power supply; they are the result of a tightly coupled design philosophy that balances aerodynamic efficiency, structural lightweighting, and robust control systems. The multidisciplinary nature of HALE UAV design is evident from the outset. Aerodynamic performance must be optimized for low Reynolds number flows, a regime where viscous effects dominate and conventional high-speed design rules no longer apply. This necessitates the use of high-aspect-ratio wings, often exceeding 20:1, to generate sufficient lift while minimizing induced drag. The wing planform is typically tapered and may incorporate winglets or blended-wing configurations to further reduce drag. Composite materials carbon-fiber reinforced polymers, glass-fiber composites, and advanced hybrid laminates are employed to achieve the required stiffness-to-weight ratios, allowing the aircraft to carry large solar arrays and energy storage systems without compromising structural integrity. The energy architecture is inherently hybrid: large, lightweight solar panels cover the wing and fuselage surfaces, feeding a battery bank that supplies power during night or low-insolation periods, while a small fuel cell or micro-turbine can provide a backup or supplementary power source during peak demand or in case of solar degradation. Finally, the control architecture must be fault-tolerant and highly redundant; small disturbances at high altitude can have amplified effects, so the aircraft typically incorporates multiple redundant actuators, sensors, and control laws that can re-route authority in the event of a failure. Despite these advances, several design gaps persist. Energy efficiency remains a critical bottleneck. Solar cells operating at high altitudes experience increased radiation and temperature extremes, which can degrade their performance over time. Moreover, the power density of current photovoltaic technologies limits the amount of energy that can be harvested per unit area, forcing designers to trade off between payload capacity and endurance. Structural integrity is another concern; the combination of low ambient pressure, large temperature swings, and the mechanical loads associated with long-duration flight can lead to material fatigue, delamination, or creep in composite structures. Existing computational fluid dynamics (CFD) tools often struggle to capture these effects accurately. Finally, stability and control systems must be designed to handle not only the nominal flight envelope but also a wide range of fault scenarios, including actuator loss, sensor drift, and unexpected atmospheric disturbances. Achieving this level of robustness requires sophisticated fault-diagnosis algorithms, adaptive control laws, and extensive redundancy, all of which add weight and complexity. The process of identifying these limitations involves a systematic review of both the literature and the design data from existing HALE platforms. By mapping out the trade-offs such as the relationship between wing loading and endurance, or between battery capacity and structural mass engineers can pinpoint where incremental improvements will yield the greatest benefit. For instance, a modest increase in solar cell efficiency can reduce the required battery capacity, thereby freeing up weight for additional payload or structural reinforcement. Similarly, adopting a new composite lay-up that offers higher modulus in the wing spars can allow for thinner wing skins, reducing drag without compromising strength. Once the gaps are identified, simulation-based validation becomes the cornerstone of the development cycle. High-fidelity CFD simulations are employed to predict aerodynamic performance across the full flight envelope, including the low Reynolds number regime. These simulations are often coupled with turbulence models that account for the unique boundary layer behavior at high altitude. Finite element analysis (FEA) is used to assess structural integrity under various loading conditions, including static loads from the solar array and dynamic loads from turbulence and control surface deflections. Energy system simulations model the interaction between solar input, battery state of charge, and power demand from avionics and propulsion, allowing designers to evaluate endurance under different weather scenarios. Control system simulations, often performed in a high-order flight dynamics environment, test the robustness of the control laws against actuator failures, sensor noise, and atmospheric disturbances. By iteratively refining the design based on these simulation results, engineers can converge on a configuration that meets the stringent endurance, payload, and reliability requirements of a solar-powered HALE UAV.

2. Market Survey and Literature review

The high-altitude long-endurance (HALE) unmanned aerial vehicle market has experienced a pronounced expansion over the past decade, driven by a confluence of civilian, military, and commercial imperatives that demand persistent, autonomous aerial platforms capable of operating at altitudes above 18 km for extended periods. The primary drivers of this growth are the escalating need for real-time situational awareness, the desire to reduce reliance on costly satellite constellations, and the increasing availability of advanced materials and power-generation technologies that make long-duration flight feasible. In the civilian sector, disaster monitoring and early-warning systems have become a cornerstone of humanitarian response, with HALE UAVs providing rapid, high-resolution imagery and atmospheric data that enable authorities to assess flood risks, wildfire spread, and the aftermath of natural catastrophes. Border surveillance and reconnaissance missions have also adopted HALE platforms to maintain continuous coverage over vast, remote frontiers, thereby enhancing national security while minimizing the operational costs associated with manned aircraft or satellite imagery. Agricultural and environmental assessment applications have leveraged the persistent presence of HALE UAVs to monitor crop health, soil moisture, and ecosystem changes over large geographic areas, providing farmers and environmental agencies with actionable insights that drive precision agriculture and conservation efforts. Perhaps the most transformative application is the use of HALE UAVs as high-altitude pseudo-satellites (HAPS), which act as communication relays, providing broadband connectivity to underserved regions, supporting disaster-response communications, and enabling low-latency data links for remote sensing and Internet-of-Things (IoT) deployments. Geographically, the most active HALE development hubs are concentrated in the United States, the United Kingdom, China, and various European nations. In the United States, the U.S. Air Force's Global Hawk program and the NASA Helios series exemplify the military and research focus on high-altitude endurance. The United Kingdom's Airbus Zephyr program, a joint venture with the U.S. Air Force, showcases a commercial-grade approach to HALE UAVs, emphasizing long-duration flight and payload versatility. China's burgeoning aerospace sector has introduced several HALE prototypes, such as the "Hengyang" series, which aim to combine solar power with advanced composite structures to achieve multi-week endurance. European initiatives, including the German Aerospace Center's (DLR) "HAPS" project and the French "Horizon" program, underscore a collaborative approach to developing reusable, persistent platforms that can serve both civilian and defense purposes. The market's maturity is evident in the commercial viability of systems such as the Zephyr, Global Hawk, Helios, and the emerging solar-powered HAPS demonstrators. These platforms have proven that HALE UAVs can deliver a cost-effective alternative to satellite deployments, offering reusable, on-demand coverage with lower launch and maintenance expenses. The demand for such platforms is particularly high in regions where satellite coverage is limited or where rapid deployment is required, such as in disaster-prone coastal zones, remote Arctic and Antarctic research stations, and expansive agricultural landscapes. Consequently, the HALE UAV market is poised for continued growth, with projected compound annual growth rates (CAGR) exceeding 10% over the next decade, driven by advancements in lightweight composites, high-efficiency photovoltaic cells, and sophisticated autonomous flight control systems.

The literature survey on HALE UAVs, provides a comprehensive overview of the aerodynamic, structural, energy, and control challenges that define the field. Several configuration studies are presented: biplane designs demonstrate comparable aerodynamic efficiency to monoplane configurations while offering higher payload capacities for the same fuel consumption, owing to their lower induced drag and inherently stiffer wing structures. Blended-wing-body (BWB) configurations exhibit improved lift-to-drag ratios and enhanced structural integrity due to the distributed load paths inherent in the joined-wing architecture. Preliminary Design of a Joined Wing Hale Uav Technical Paper explores joined-wing configurations that achieve a span efficiency factor of 1.4–1.5, thereby reducing induced drag and structural weight relative to conventional wing-body designs. These aerodynamic insights underscore the importance of exploring unconventional airframe geometries to maximize endurance and payload while minimizing power consumption. Structural design insights from the survey emphasize the extensive use of carbon-fiber-reinforced polymer (CFRP) composites, which provide the necessary stiffness-to-weight ratios for high-aspect-ratio wings and large solar-array panels. Key structural considerations include weight minimization, flutter margins, and reliability and redundancy. Manufacturing processes such as autoclave curing and resin-transfer molding are highlighted as critical to achieving the required material properties and dimensional tolerances. The literature also points to the need for rigorous fatigue testing under realistic atmospheric turbulence conditions, as long-duration flight exposes the structure to repeated load cycles that can lead to crack initiation and propagation over time. Energy system analyses in the survey compare lithium-ion batteries and hydrogen fuel cells. The findings indicate that fuel cells outperform batteries when the energy demand exceeds 2.8 kWh, a threshold that is often surpassed during high-altitude operations due to increased power demand

(3.5% per km altitude). Thermal management emerges as a pivotal factor, as both battery and fuel-cell performance are highly temperature-dependent. Solar cell studies, supporting research, reveal that triple-junction silicon cells achieve efficiencies between 25% and 30%, while thin-film cells offer lower efficiencies but benefit from reduced weight and flexibility. Night-time endurance is identified as the most critical limitation, primarily due to battery degradation and storage inefficiencies that reduce the available energy reserve during periods of low solar irradiance. Stability and control findings emphasize the necessity of longitudinal and directional stability for sustained flight. Flutter analysis is critical for high-aspect-ratio wings, as aeroelastic coupling can lead to catastrophic failure if not properly mitigated. Control surface sizing is linked to low-density air performance, requiring larger deflection angles or higher actuation authority to achieve the same control effectiveness as at lower altitudes. The literature also highlights the importance of fault-tolerant control architectures that can maintain stability in the event of actuator or sensor failures, given the high-altitude operational envelope and the limited opportunities for ground intervention. The survey identifies several gaps that warrant further investigation.

- First, the integration of hybrid energy systems combining solar, battery, and fuel-cell technologies remains limited, with few studies exploring optimal sizing and control strategies for these systems under varying mission profiles.
- Second, practical design considerations under real atmospheric turbulence, including turbulence intensity, gust loads, and wind shear, are underexplored, particularly in the context of long-duration flight where cumulative effects can degrade structural integrity and control performance.
- Third, long-duration structural fatigue data are scarce, hindering the ability to predict service life and schedule maintenance for HALE platforms.

Finally, optimized battery-fuel-cell combinations, including the use of advanced chemistries and hybrid power-management strategies, are still underexplored, leaving a significant opportunity to enhance endurance and reduce weight. By synthesizing these findings, the literature review underscores the multifaceted challenges that must be addressed to advance HALE UAV technology. It also highlights the critical need for interdisciplinary research that couples aerodynamic optimization, advanced composite design, hybrid energy system integration, and robust control strategies to realize the full potential of high-altitude, long-endurance unmanned platforms.

3. Aim and Objectives

3.1 Aim

The overarching aim of this research is to conceive, analyze, and validate a theoretically sound design for a solar-powered High-Altitude Long-Endurance (HALE) Unmanned Aerial Vehicle (UAV). The design will be grounded in rigorous aerodynamic, structural, and power-system analyses, and will be corroborated through high-fidelity simulation tools. By integrating state-of-the-art solar photovoltaics, advanced energy storage, and lightweight composite structures, the proposed HALE platform will demonstrate the feasibility of achieving multi-week endurance while maintaining payload versatility and operational reliability.

3.2 Objectives

Primary Objective

Conceptual Design Formulation and Validation: Develop a comprehensive conceptual design of a solar-powered HALE UAV that satisfies mission endurance, payload, and safety requirements. This includes defining the airframe geometry, wing planform, propulsion architecture, and power-management strategy. The design will be validated through a combination of analytical calculations, computational fluid dynamics (CFD), finite element analysis (FEA), and system-level simulations to ensure that all subsystems meet performance targets under realistic environmental conditions.

Secondary Objectives

1. In-Depth Literature Survey

- Conduct a systematic review of existing solar UAVs and HALE system designs, focusing on aerodynamic configurations (biplane, blended-wing, joined-wing), structural materials (CFRP, hybrid laminates), and energy-storage technologies (lithium-ion, hydrogen fuel cells, supercapacitors).

- Synthesize findings to identify best practices, emerging trends, and gaps in current knowledge that can inform the proposed design.

2. Identification of Design Limitations

- Critically assess the limitations of current energy-storage solutions, particularly battery degradation, thermal management, and weight penalties.
- Evaluate aerodynamic challenges such as low-Reynolds-number performance, induced drag, and flutter susceptibility in high-aspect-ratio wings.
- Examine structural constraints, including fatigue life under long-duration turbulence, material degradation at high altitudes, and the trade-off between stiffness and weight.

3. Simulation-Based Performance Evaluation

- Employ CFD to model airflow over the proposed wing and fuselage geometries, capturing low-Reynolds-number effects and validating lift-to-drag ratios.
- Use FEA to assess structural integrity, modal characteristics, and fatigue life under realistic load spectra.
- Integrate power-system simulations to predict endurance, energy budgets, and mission reliability under varying atmospheric conditions.

4. Propose Design Improvements and Future Research Directions

- Based on simulation outcomes, recommend specific design modifications such as optimized wing-tip shapes, advanced composite lay-ups, or hybrid power-management strategies to enhance performance.
- Outline a roadmap for future experimental validation, including prototype fabrication, wind-tunnel testing, and flight trials.
- Identify key research questions that remain open, such as the integration of next-generation photovoltaic technologies or the development of autonomous fault-diagnosis algorithms for long-endurance missions.

5. Renewable Energy Integration Analysis

- Quantify the impact of renewable energy sources on overall endurance and mission reliability by modeling energy generation profiles across diurnal cycles and varying solar irradiance.
- Investigate the trade-offs between solar array area, battery capacity, and fuel-cell sizing to achieve optimal weight distribution and power balance.
- Assess the resilience of the energy system to extreme events (e.g., solar flares, high-altitude turbulence) and propose mitigation strategies such as redundant power paths or adaptive power-management schemes.

By systematically addressing these objectives, the research will deliver a robust, theoretically validated design for a solar-powered HALE UAV, while also charting a clear path for future experimental work and technological advancement in the field.

4. Problem Definition and Design Specification

The design challenge is to synthesize a solar-powered High-Altitude Long-Endurance (HALE) Unmanned Aerial Vehicle (UAV) that can sustain continuous flight above 18 km for a minimum of 24 hours, with the ambition of extending that endurance to several days. Achieving this objective demands a tightly coupled integration of three core subsystems **aerodynamics, structure, and energy** that each must meet or exceed the performance envelopes.

Conceptual performance results:

Cruise power ≈ 2 kW;
Solar Power available ≈ 9 kW;
Night Storage ≈ 24 kWh;
Battery Mass ≈ 95 kg;
Hydrogen equivalent ≈ 1.44 kg (≈ 4.3 kg system)

Aerodynamic Integration:

At stratospheric altitudes the air density drops to roughly one-tenth of sea-level values, forcing the aircraft to operate at low Reynolds numbers (10^5 – 10^6). Conventional airfoils lose lift and suffer from increased drag in this

regime, so the design must adopt a wing configuration that maximizes lift-to-drag (L/D) while minimizing induced drag. Three promising topologies emerge: a biplane arrangement that effectively doubles the aspect ratio without a proportional increase in span, a blended-wing-body (BWB) that merges fuselage and wing into a single lifting surface, and a joined-wing that eliminates the traditional fuselage altogether, thereby reducing structural bending moments and improving span efficiency by 40–50 %. Each configuration will be evaluated using high-fidelity CFD at representative altitudes (18–22 km) and flight speeds ($\approx 70\text{--}90\text{ km h}^{-1}$) to quantify lift, drag, and pressure distributions. The selected airfoil family will be optimized for low-Reynolds-number performance, targeting a maximum lift coefficient ($C_{l,\text{max}}$) of at least 2.5 and a drag polar slope that keeps parasitic drag below 0.02 at cruise Mach numbers.

Structural Optimization:

The structural design must reconcile the need for an ultra-light, high-stiffness airframe with the demands of flutter safety and fatigue resistance. Carbon-fiber-reinforced polymer (CFRP) composites, with their exceptional stiffness-to-weight ratio, will form the primary load-carrying elements. A modular wing-box construction will allow for pre-stressing of spars and ribs to counteract the high lift loads at altitude. Finite-element analysis (FEA) will be employed to predict static deflection, modal frequencies, and flutter margins under worst-case loading scenarios (e.g., gusts of 5 m s^{-1} at 20 km). The design will target a maximum wing deflection of less than 5 % of the chord and a flutter speed that exceeds the cruise Mach number by at least 30 %. Additionally, the fuselage will be engineered as a semi-monocoque structure with integrated payload bays, ensuring that the center of gravity remains within $\pm 5\%$ of the neutral point throughout the mission envelope.

Hybrid Energy System:

Solar power is the primary energy source during daylight, but the intermittent nature of sunlight and the high energy demands of night-time flight necessitate a hybrid storage strategy. Triple-junction gallium arsenide (GaAs) cells, delivering 25–30 % efficiency under 1 kW m^{-2} irradiance, will be mounted on the wing and fuselage surfaces to maximize available area while keeping mass low. Thin-film silicon or organic cells will be reserved for flexible or curved surfaces where weight penalties are prohibitive. Excess solar energy will be stored in high-cycle lithium-ion batteries ($\approx 250\text{ Wh kg}^{-1}$) that can deliver the required power during low-irradiance periods. For missions exceeding 2.8 kWh of night-time demand, a hydrogen fuel cell stack ($\approx 0.5\text{ kg kWh}^{-1}$) will be incorporated, powered by a lightweight on-board electrolyzer that recharges the hydrogen tank during daylight. The energy management system will be modeled using a mixed-integer linear program (MILP) to schedule power flows, ensuring that the battery state of charge never falls below 20 % and that the fuel cell operates within its optimal power band.

Mission Profile and Constraints:

The aircraft will cruise at 18–22 km, maintaining a low, steady speed ($\approx 80\text{ km h}^{-1}$) to maximize aerodynamic efficiency. Payload capacity will be limited to 5–10 kg, sufficient for scientific instruments or communication payloads. Environmental constraints such as cloud cover, temperature gradients, and wind shear will be incorporated into the mission simulator to assess robustness. Weight and balance limits will be enforced through a mass-balance analysis that accounts for fuel, batteries, payload, and solar array mass distribution.

Design Validation:

The final design will be validated through a combination of analytical calculations, CFD, FEA, and system-level simulations. Key performance indicators lift-to-drag ratio, structural flutter margin, energy balance, and endurance will be benchmarked against the literature-derived targets. The outcome will be a theoretically sound, end-to-end HALE UAV concept that demonstrates feasibility for sustained, high-altitude, long-endurance flight, thereby laying the groundwork for future experimental prototypes and operational deployments.

5. Methodology:

5.1 Literature-Based Foundation:

The initial phase of the project was anchored in an exhaustive review of the state-of-the-art research that pertains to high-altitude, long-endurance (HALE) unmanned aerial vehicles. By systematically scanning peer-reviewed journals, technical reports, and relevant patents, compiled a comprehensive knowledge base that covered every major discipline influencing HALE performance. This effort yielded a catalog of existing HALE platforms such as the NASA Helios, Airbus Zephyr, and the DARPA Vulture together with their specific mission parameters (cruise altitude, endurance, payload, and power budgets). Parallel to the aerodynamic survey, the literature on solar-cell technologies was examined, focusing on the relative merits of triple-junction versus thin-film devices in terms of specific power (W/kg), radiation tolerance, and temperature coefficients at stratospheric conditions. Energy-storage options were also scrutinized, with a balanced assessment of lithium-ion batteries (energy density, cycle life, thermal management) and fuel-cell systems (specific power, fuel logistics, degradation mechanisms). Structural and material constraints were identified by reviewing studies on high-aspect-ratio wing flexure, composite lay-up optimization, and fatigue life under cyclic loading. Finally, the review synthesized the prevailing stability, control, and flutter requirements that have been validated in flight tests, thereby exposing the technical gaps that the present design must address.

5.2 Mission Requirements Definition:

Building on the insights gathered from the literature, a concrete set of mission requirements was distilled to serve as the design envelope for the conceptual HALE UAV. Endurance is a critical metric; the UAV must sustain continuous flight for a minimum of 24 hours, which necessitates a reliable day-night energy balance. The primary energy source is solar power harvested during daylight, complemented by an on-board storage system either batteries, fuel cells, or a hybrid combination to provide power through the night. A target lift-to-drag (L/D) ratio is required to ensure that the aircraft can cruise at the prescribed altitude with minimal propulsive power; preliminary calculations suggest an L/D of 25–30 is necessary to meet the endurance goal. Payload capacity must be defined in terms of both mass and volume, allowing for scientific instruments, communications equipment, or surveillance sensors while preserving the overall weight budget. Structural stiffness is another non-negotiable requirement; the wing must retain sufficient torsional rigidity to avoid flutter across the full flight envelope, especially given the high aspect ratios typical of HALE designs. Finally, a comprehensive weight budget was established, allocating mass to the airframe, propulsion, energy-storage, avionics, and payload, and ensuring that the sum does not exceed the lift capability dictated by the aerodynamic analysis. These constraints collectively shape every subsequent modeling, simulation, and design decision.

5.3 Aerodynamic Modelling:

5.3.1 Airfoil and Wing Analysis

The aerodynamic evaluation proceeds through a two-tiered approach that couples analytical theory with high-fidelity computational fluid dynamics (CFD). First, candidate airfoils are screened using established low-Reynolds-number performance data, focusing on lift coefficient (C_L), drag coefficient (C_D), and the resulting C_L/C_D ratio across the expected operating Reynolds range ($\approx 0.5\text{--}2 \times 10^6$). Stall characteristics are examined by plotting polar curves and identifying the angle of attack at which lift begins to deteriorate sharply, ensuring sufficient margin for gust tolerance. The influence of Reynolds number on both laminar-to-turbulent transition and boundary-layer thickness is quantified, as these effects become pronounced at stratospheric altitudes. Once a shortlist of airfoils is identified, CFD simulations are conducted on representative wing sections to capture three-dimensional effects such as tip vortices, spanwise flow, and pressure-gradient induced separation. The results are validated against wind-tunnel data where available, and the final airfoil selection balances high lift, low drag, and robust stall behavior under the low-density conditions of HALE flight.

5.3.2 Wing Configuration Comparison

Three distinct wing configurations are examined biplane, blended-wing body (BWB), and joined-wing each offering a unique set of aerodynamic and structural benefits. The biplane concept, while historically associated with higher induced drag, can achieve a lower effective aspect ratio through constructive interference of the two lifting surfaces, potentially reducing wing loading and improving low-speed handling. The BWB configuration provides a smooth pressure distribution across the fuselage-wing junction, which translates into higher overall L/D ratios and

a more efficient use of structural material due to the continuous load path. The joined-wing arrangement, characterized by intersecting wing panels that form a “X” shape, yields a span efficiency factor (e) in the range of 1.4–1.5, thereby reducing induced drag and distributing bending moments more evenly along the span. For each configuration, Prandtl’s lifting-line theory is applied to estimate the induced drag component, while aspect ratio (AR) and span efficiency are incorporated into the drag polar. Mission-specific lift requirements—derived from the weight budget and cruise altitude—are then used to size the wings, ensuring that the required lift coefficient can be achieved without exceeding structural limits. The comparative analysis ultimately identifies the configuration that best satisfies the trade-off between aerodynamic efficiency, structural weight, and manufacturability for the HALE mission.

5.4 Structural Modelling:

5.4.1 Material Selection Strategy

Carbon-fiber-reinforced polymer (CFRP) was selected as the baseline structural material because it offers an unparalleled combination of high specific strength, low density, and excellent fatigue resistance attributes that are essential for a wing with an aspect ratio exceeding 20. The material’s anisotropic nature allows the designer to tailor stiffness and strength by orienting the fiber plies in directions that align with the principal load paths, thereby minimizing unnecessary mass. Moreover, recent advances in out-of-plane reinforcement and nano-engineered resin systems further improve damage tolerance and reduce the risk of delamination under cyclic loading. These characteristics directly address the HALE structural challenges identified in the literature review, such as maintaining wing integrity over long-duration flights and resisting flutter at high altitudes.

5.4.2 Structural Analysis

A detailed finite-element model of the airframe is constructed using ANSYS Workbench, with geometry generated in CATIA to capture the exact curvature and thickness distribution of the wing, fuselage, and control surfaces. The model incorporates realistic boundary conditions, including aerodynamic pressure loads derived from the CFD results and inertial loads from the flight dynamics simulation. Primary structural metrics wing bending stress, root shear, and torsional stiffness are extracted from the FEM solution, and the results are compared against material allowable stresses and buckling criteria. A dedicated flutter analysis is performed using the coupled aero-elastic module, which integrates the aerodynamic influence coefficients (computed via a doublet-lattice method) with the structural modal data to predict the critical flight speed at which flutter may occur. Deformation patterns under cruise and gust loads are visualized to verify that the wing deflection remains within acceptable limits, preserving aerodynamic performance and control authority. The outcome of this analysis informs iterative refinements of the lay-up schedule, spar geometry, and skin thickness, ensuring that the final structural design can safely withstand the operational loads of a HALE mission.

5.5 Energy System Modelling

5.5.1 Solar Power Estimation:

The solar-energy subsystem is sized based on the performance of triple-junction photovoltaic cells, which currently achieve specific powers in excess of 350 W/kg under stratospheric irradiance ($\approx 1,360 \text{ W/m}^2$). Using the available wing surface area after accounting for structural ribs, control surfaces, and necessary shielding the maximum solar-collection area is calculated. The model incorporates the cosine loss due to the angle of incidence between the sun vector and the panel normal, as well as shading effects from wing-tip devices and any onboard antennas. Seasonal and diurnal variations in solar flux at 18–20 km altitude are also considered, allowing the estimation of daily energy harvest curves. The resulting power generation profile serves as the primary input for the energy-balance analysis, indicating the surplus energy available for storage during daylight hours.

5.5.2 Battery / Fuel-Cell Simulation

Given the literature finding that fuel cells become more efficient than batteries when the required nightly energy exceeds roughly 2.8 kWh, a hybrid storage architecture is modeled in MATLAB/Simulink. The battery subsystem is represented by a lithium-ion pack with a defined specific energy ($\approx 250 \text{ Wh/kg}$), charge-discharge efficiency, and degradation curve that captures capacity loss over the mission’s expected cycle count. The fuel-cell subsystem is modeled using a proton-exchange-membrane (PEM) stack, with performance maps that relate output power to

hydrogen flow rate, temperature, and pressure. The simulation tracks the instantaneous power demand of the propulsion and avionics loads, the solar input, and the state of charge (SoC) of each storage element. Night-time endurance is evaluated by integrating the net power deficit over the dark period, and sensitivity analyses explore how variations in temperature, solar angle, and component aging affect the ability to meet the 24-hour endurance target.

5.5.3 Hybrid Energy Management Framework

An overarching energy-balance model is constructed to orchestrate the flow of power from the solar array to the storage devices and ultimately to the propulsion and avionics subsystems. The framework employs a rule-based controller that prioritizes charging the battery when excess solar power is available, while allocating any remaining surplus to the fuel-cell system for hydrogen production (if on-board reforming is considered) or for direct power supplementation. During night, the controller draws first from the battery until its SoC reaches a predefined safety threshold, after which the fuel cell takes over to sustain the remaining power demand. This hierarchical management strategy is simulated over multiple diurnal cycles to verify that the UAV can maintain continuous operation without violating any component's operational limits. The model also quantifies the trade-off between solar-panel area (and thus wing weight) and the required storage mass, providing a quantitative basis for the subsequent design optimization.

5.5.4 Numerical Validation Using Conceptual HALE Model

Using the HALE geometric model shown (main wing chord 1 m, span 32 m, tail semi-span 2.5 m, structural mass distribution as given), a complete conceptual energy-aerodynamic performance estimate was computed. The results show that the model requires approximately 1.35 kW propulsive power at cruise, translating to 1.99 kW of electrical demand after system losses. With 30.4 m² of solar-covered area, the UAV can generate ~9.38 kW of usable solar power at 18–20 km altitude, proving that the platform has strong daytime power surplus. Night-time energy demand for a 12-hour cycle was found to be ~23.9 kWh, requiring ~95.7 kg of lithium-ion batteries, but only ~1.44 kg of hydrogen (~4.31 kg including tank and system mass). These results confirm that a pure-battery system is infeasible for HALE endurance, whereas a solar and hydrogen fuel-cell hybrid model is mass-efficient.

5.6 Stability and Control Analysis:

The stability assessment builds on classical longitudinal and directional stability theory, adapted for the low-density environment of HALE flight. Static margin is calculated by locating the aerodynamic center of the wing-body configuration and comparing it to the center of gravity (CG) location, ensuring a margin of at least 5 % of the mean aerodynamic chord to guarantee positive static stability. Control surface sizing elevators, rudders, and ailerons is performed using the derivative method, where the required hinge moments are derived from the desired control authority at cruise Mach numbers ($\approx 0.2\text{--}0.3$). The analysis also incorporates the effect of reduced dynamic pressure on control effectiveness, prompting the inclusion of larger surface areas or the use of high-deflection actuators. Flutter boundaries are estimated using the Doublet Lattice Method (DLM) to generate aerodynamic influence coefficient matrices, which are then coupled with the structural modal data from the FEM analysis. By sweeping flight speed and altitude, the critical flutter speed is identified, and the design is iteratively refined (e.g., by increasing torsional stiffness or adjusting mass distribution) to ensure that the operational envelope remains comfortably below this limit.

5.7 Performance Simulation:

A suite of integrated simulations is executed to verify that the conceptual UAV meets all performance targets. Aerodynamic performance is evaluated by feeding the CFD-derived lift and drag polar into a mission-profile script that computes the required thrust, fuel (or battery) consumption, and resulting L/D ratio across the flight envelope. Structural performance is examined by extracting stress and deformation results from the FEM model under the combined aerodynamic and inertial loads, confirming that all stresses stay within allowable limits and that deflections do not degrade aerodynamic efficiency. Energy performance is assessed by coupling the solar-power model with the hybrid storage simulation, producing time-resolved power-balance plots that show the net surplus or deficit throughout a 24-hour cycle. An altitude-power relationship is also generated, incorporating the empirically observed 3.5 % increase in power demand per kilometre of altitude gain; this relationship informs the

selection of cruise altitude that balances reduced drag against higher propulsion power requirements. All simulations are performed in MATLAB for the system-level analyses and ANSYS for the high-fidelity aero-structural coupling, ensuring consistency across the multidisciplinary model.

5.8 Design Integration and Optimization:

The final design stage brings together the aerodynamic, structural, and energy subsystems into a single, tightly coupled optimization loop. A multi-objective genetic algorithm (MOGA) is employed to explore the design space, simultaneously minimizing total mass, maximizing endurance, and maximizing L/D while satisfying constraints on flutter margin, structural stress, and solar-panel area. Trade-offs are explicitly quantified: increasing solar-panel area improves daytime power generation but adds wing weight and may reduce aspect ratio; adding battery mass extends night-time endurance but also raises wing loading; raising aspect ratio improves aerodynamic efficiency but heightens flutter susceptibility. The optimizer iteratively adjusts variables such as wing span, chord distribution, spar geometry, CFRP lay-up sequence, solar-cell layout, and storage sizing. Convergence is achieved when successive generations produce negligible improvements in the objective functions, and the resulting design satisfies all mission requirements with a comfortable safety margin.

5.9 Validation Against Literature Benchmarks

The culminating step is a rigorous validation of the conceptual UAV against established benchmarks from the HALE literature. The predicted endurance (≥ 24 h) is compared with recorded flight times of platforms like the Zephyr (up to 25 days) and the Helios (up to 10h), confirming that the design is competitive. Aerodynamic performance specifically the L/D ratio and aspect ratio is cross-checked against published data for similar wing configurations, ensuring that the simulated values fall within the 5-10% tolerance typical of validated CFD studies. Energy feasibility is evaluated by juxtaposing the modeled solar-to-storage balance with real-world measurements from solar-powered UAV testbeds, verifying that the hybrid system can indeed sustain day-night operation under realistic irradiance and temperature conditions. Finally, structural feasibility is corroborated by comparing the FEM-derived stress distribution and flutter margin with those reported for CFRP-based HALE wings in peer-reviewed studies. This comprehensive benchmarking exercise confirms that the conceptual design not only aligns with scientific expectations but also offers a credible pathway toward a flight-worthy HALE UAV.

6. Requirements / Resources

The successful execution of this project requires a combination of technical resources, computational tools, research materials, and academic guidance. Since the design of a HALE UAV is inherently multidisciplinary, the work depends heavily on access to comprehensive literature databases, simulation software, modelling environments, and expert inputs. To establish the theoretical foundation, academic resources such as technical reports, NASA archives, and aerospace design publications are essential. These sources provide validated aerodynamic data, structural analysis methods, and energy system performance characteristics necessary to define design constraints and benchmark the conceptual UAV against existing platforms. The documents uploaded for this project already synthesize a wide range of such references, ensuring continuity in research direction and design methodology.

From a computational perspective, advanced engineering software forms the core of the design and simulation workflow. MATLAB and Simulink are required for developing performance models, analysing energy systems, and simulating battery and fuel cell behaviour under varying atmospheric conditions. These tools also support system-level modelling, enabling the integration of solar input, energy storage, and propulsion requirements. Aerodynamic and structural analyses require high-fidelity simulation environments such as ANSYS for CFD and FEM studies, allowing the evaluation of lift, drag, stress distribution, deformation, and structural stability. CATIA is required for accurate 3D modelling of the airframe, wing configurations, and component layouts, supporting both geometric design and downstream analysis tasks. These tools were also highlighted in the PPT as essential simulation resources for the project's technical. In addition to software, expert guidance is a crucial resource. Consultations with faculty members, aerospace engineers, and subject experts support the validation of assumptions, interpretation of simulation outputs, and refinement of design strategies. This ensures that the

conceptual model aligns with established aerospace practices and adheres to realistic engineering constraints. Furthermore, atmospheric data, solar irradiance profiles, and material property datasets are necessary to accurately model real-world operating conditions at altitudes above 18 km conditions that significantly influence aerodynamic and energy performance.

Finally, project management resources such as a structured timeline, iterative documentation, and access to laboratory computing facilities are required to ensure continuity and systematic progress. Collectively, these resources enable comprehensive analytical development, modelling accuracy, and reliable assessment of the feasibility of the proposed solar-powered HALE UAV design.

7. Conclusion and Future Work

The study undertaken in this project demonstrates the technical feasibility and multidisciplinary complexity involved in the conceptual design of a High Altitude Long Endurance (HALE) Unmanned Aerial Vehicle powered primarily by renewable energy. Through an in-depth literature review, it was established that HALE UAVs rely on a delicate integration of aerodynamics, lightweight structural materials, and high-efficiency energy systems to achieve sustained flight at altitudes exceeding 18 km. Existing research highlights key technological limitations, particularly concerning night-time endurance, structural stability of high-aspect-ratio wings, and the efficiency of hybrid energy storage systems. By synthesizing this information, the project successfully defined a clear engineering problem statement and developed rigorous design specifications that align with documented performance benchmarks for HALE platforms. The methodology employed spanning mission requirement analysis, aerodynamic modelling, structural evaluation, and energy system simulations provides a structured foundation for evaluating a solar-powered HALE UAV concept. The use of tools such as MATLAB, ANSYS, and CATIA supports the quantitative assessment of aerodynamic behaviour, structural load distribution, and energy availability across a full diurnal cycle, as highlighted in the project's initial framework. Although the project remains theoretical, the findings emphasize that a carefully optimized combination of triple-junction solar cells, composite airframe structures, and hybrid battery–fuel cell systems can potentially achieve continuous multi-hour to multi-day endurance under suitable environmental conditions. However, several challenges remain and form the basis for future work. The foremost requirement is the development of detailed computational models that incorporate real atmospheric variations, including turbulence, temperature gradients, and cloud shading, all of which significantly influence HALE performance. Prototype-scale CFD studies and wind tunnel validation are essential to refine aerodynamic characteristics, particularly under low Reynolds number conditions typical of high-altitude flight. Structural testing of composite wing segments is equally important to investigate flutter boundaries and ensure material reliability during extended missions. Further, a dedicated energy management algorithm must be developed to optimally allocate power between solar input, battery storage, and fuel cell output, especially during transitional periods such as dawn, dusk, or rapidly changing weather. The next stage of the project should involve the fabrication of a scaled demonstrator UAV to validate theoretical predictions through flight testing. This will enable the measurement of real-world performance metrics, such as lift-to-drag ratios, power consumption, solar charging efficiency, and stability characteristics. Additional research into emerging technologies such as morphing wings, solid-state batteries, and ultra-light flexible solar arrays can contribute to further improvements in endurance and structural efficiency. The project provides a strong conceptual and analytical groundwork for the development of a renewable-energy-powered HALE UAV. With continued refinement, simulation, and experimental validation, the design has the potential to evolve into a viable platform for long-endurance atmospheric monitoring, communication support, and environmental data acquisition. The numerical evaluation of the conceptual HALE model confirms that the aircraft geometry and aerodynamic assumptions produce realistic power demands (~2 kW electrical), while the high-altitude solar flux combined with triple-junction photovoltaics provides ~9 kW of usable power, ensuring robust daytime performance. The analysis also quantifies night-time storage requirements, showing that pure lithium-ion battery solutions exceed feasible mass limits (~95 kg), whereas hydrogen fuel-cell storage meets the requirement at <5 kg system mass. These results reinforce the feasibility of a renewable-energy-powered HALE UAV when hybrid energy storage solutions are adopted.

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